Lecture 10

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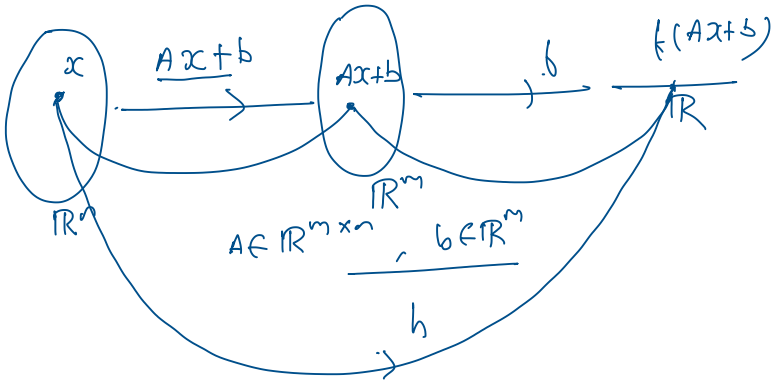
Recap
- Convex sets
- Convex / strictly convex / strongly convex functions
- Concave / " concave / " concave functions

Example $f: \mathbb{R}^n \longrightarrow \mathbb{R}$
$f(x) = \|x\|$
- convex but not strictly convex.

Today $f: \mathbb{R}^m \longrightarrow \mathbb{R}$, convex.



Define $h: \mathbb{R}^n \longrightarrow \mathbb{R}$
$h(x) = f(Ax+b)$

claim: if f is convex, so is h.

$x, y \in \mathbb{R}^n$, $h(\alpha x + (1-\alpha) y) = f(A(\alpha x + (1-\alpha) y) + b)$

$= f(\alpha (Ax+b) + (1-\alpha)(Ay+b))$.

convexity of f, $\leq \alpha f(Ax+b) + (1-\alpha) f(Ay+b)$.

$= \alpha h(x) + (1-\alpha) h(y)$ ∎

[Equation] , [Equation] . Define [Equation] as [Equation]

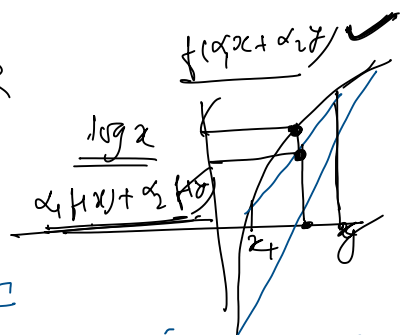
- If f a convex/concave function so is [Equation] .

[Equation] is a convex set

Example ① $f(x) = \|Ax+b\|$ is convex.

$g(x) = \|x\|$ is convex

② $g: \{x : a^T x + b > 0\} \longrightarrow \mathbb{R}$
$g(x) = \log(a^T x + b)$
is concave.



Convexity $f: S \longrightarrow \mathbb{R}$, a convex fn

$f(\alpha_1 x + \alpha_2 y) \leq \alpha_1 f(x) + \alpha_2 f(y)$ $\alpha_1, \alpha_2 \geq 0$ and $\alpha_1 + \alpha_2 = 1$

Now consider $x_1, x_2, x_3 \in S$
$\alpha_1, \alpha_2, \alpha_3 \geq 0$ $\sum_{i=1}^{3} \alpha_i = 1$

$f(\alpha_1 x_1 + \alpha_2 x_2 + (\alpha_1 + \alpha_2))$

$$f(\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3) = f\left(\alpha_1 x_1 + (\alpha_2+\alpha_3)\left(\frac{\alpha_2 x_2 + \alpha_3 x_3}{\alpha_2+\alpha_3}\right)\right)$$

convexity of f

$$\leq \alpha_1 f(x_1) + (\alpha_2+\alpha_3) f\left(\frac{\alpha_2 x_2 + \alpha_3 x_3}{\alpha_2+\alpha_3}\right)$$

$$\leq \alpha_1 f(x_1) + \cancel{(\alpha_2+\alpha_3)} \frac{\alpha_2}{\cancel{\alpha_2+\alpha_3}} f(x_2) + \frac{\alpha_3}{\cancel{\alpha_2+\alpha_3}} f(x_3)$$

$$= \alpha_1 f(x_1) + \alpha_2 f(x_2) + \alpha_3 f(x_3)$$

- If [Equation] is a ~~convex~~ function then [Equation] for all [Equation] and [Equation] such that [Equation] .

Example X is a discrete R.V., takes values $x_1, x_2, \ldots, x_n$ w.p. $p_1, p_2, \ldots, p_n$

$p_i \in [0,1]$, $\sum_{i=1}^{n} p_i = 1$, $x_i \in S \ \forall i$

S convex set

Consider $f: S \to \mathbb{R}$, convex

$$f\left(\sum p_i x_i\right) \leq \sum p_i f(x_i)$$

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$ Jensen's inequality

The inequality also holds when X is a CR R.V.

Suppose X has pmf $p(x)$

$$f\left(\int x\, p(x)\, dx\right) \leq \int p(x) f(x)\, dx$$

[Equation] is a convex set, [Equation] and [Equation] is a random variable.

- If [Equation] is a convex then [Equation] (Jensen's inequality)
- If [Equation] is a concave then [Equation]

Examples ① $f(x) = x^2$ $(f: \mathbb{R} \to \mathbb{R})$

– a convex fn

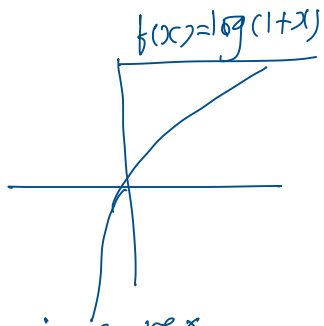
$$\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$$

$$\Rightarrow \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \geq 0$$

i.e. $\mathrm{Var}(X) \geq 0$

② $\mathbb{E}[\log(1+X)] \leq \log \mathbb{E}[(1+X)]$

$= \log(1 + \mathbb{E}[X])$



$\alpha_1, \alpha_2 > 0$, f is convex.

$f(\alpha_1 x_1 + \alpha_2 x_2)$ $\alpha_1 f(x_1) + \alpha_2 f(x_2)$

$$\text{L.H.S} = f\left(\frac{\alpha_1}{\alpha_1+\alpha_2}(\alpha_1+\alpha_2)x_1 + \frac{\alpha_2}{\alpha_1+\alpha_2}(\alpha_1+\alpha_2)x_2\right)$$

$$\leq \frac{\alpha_1}{\alpha_1+\alpha_2} f((\alpha_1+\alpha_2)x_1) + \frac{\alpha_2}{\alpha_1+\alpha_2} f((\alpha_1+\alpha_2)x_2)$$

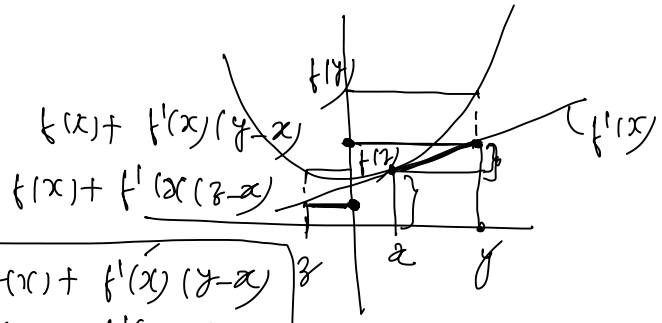
$$\overset{?}{\leq} \alpha_1 f(x_1) + \alpha_2 f(x_2)$$

$f(x) = x^2$ $\quad \alpha_1 = 100, \ \alpha_2 = 100, \ x_1 = 1, \ x_2 = 2$

$\text{L.H.S} = f(300) = 9 \times 10^4$

$\text{R.H.S} \ \alpha_1 f(x_1) + \alpha_2 f(x_2) = 100 \times 1 + 100 \times 4 < \text{L.H.S}$

(desired inequality does not hold)



$$\boxed{\begin{aligned} f(y) &\geq f(x) + f'(x)(y-x) \\ f(z) &\geq f(x) + f'(x)(z-x) \end{aligned}}$$

First order characterization ([Equation] convex, $f: S \to \mathbb{R}$ differentiable):

- [Equation] is convex iff
  [Equation] for all $x, y \in S$
- [Equation] is strictly convex iff
  [Equation] for all [Equation]
- [Equation] is strongly convex with parameter $m > 0$ iff
  [Equation] for all $x, y \in S$

Proof of the first part: suppose f is convex.

$$\Rightarrow f(\alpha x + (1-\alpha) y) \leq \alpha f(x) + (1-\alpha) f(y)$$

$$\Rightarrow f(\alpha x + (1-\alpha) y) - f(x) \leq (1-\alpha)(f(y) - f(x))$$

$$\Rightarrow \frac{f(x + (1-\alpha)(y-x)) - f(x)}{1-\alpha} \leq f(y) - f(x)$$

take $\alpha \to 1$ (i.e. $(1-\alpha) \to 0$) on L.H.S

$$\nabla f(x)^T (y-x) \leq f(y) - f(x)$$

$$\Rightarrow f(y) \geq f(x) + \nabla f(x)^T (y-x)$$

Second order characterization ([Equation] convex, $f: S \to \mathbb{R}$ twice differentiable):

- [Equation] is convex iff
  [Equation] ($\nabla^2 f(x)$ is positive semidefinite) all $x \in S$
- [Equation] is strictly convex if
  [Equation] ($\nabla^2 f(x)$ is positive definite) all $x \in S$
- [Equation] is strongly convex with parameter $m > 0$ iff
  [Equation] all $x \in S$

$\nabla^2 f(x)$ is PD $\Rightarrow$ f is strictly convex.

f is strictly convex $\not\Rightarrow$ $\nabla^2 f(x)$ is PD

$\nabla^2 f(x) - mI$ is PSD

$\forall\ \lambda_{min}(\nabla^2 f(x) - mI) \geq 0$

$\forall\ \lambda_{min}(\nabla^2 f(x)) \geq m$

$m + \lambda(A) = \lambda(A + mI)$

(If f is scalar, $f''(x) - m \geq 0$)

Examples ① $f(x) = x^4$ is strictly convex.

$f'(x) = 4x^3$

$f''(x) = 12x^2 \not> 0 \quad \forall x \in \mathbb{R}$

(fails at $x=0$)

Is f strongly convex?

Is there an $m > 0$ s.t.

$f''(x) - m \geq 0 \quad \forall x$

So f is not strongly convex.

② $f: [2, 3] \to \mathbb{R}$

$f(x) = x^4$

$f''(x) = 12x^2$

$12x^2 - m \geq 0 \quad \forall x \in [2,3]$

satisfied for $m = 48$

- f is strongly convex with parameter 48.

③ $A \in \mathbb{R}^{n \times n}$ symmetric, $b \in \mathbb{R}^n$

$f(x) = \frac{1}{2} x^T A x + bx + c$

$\nabla f(x) = Ax + b$

$\nabla^2 f(x) = A$

f is convex if $\lambda_{min}(A) \geq 0$

f is strictly convex if $\lambda_{min}(A) > 0$

- in this case f is also strongly convex with parameter $\lambda_{min}(A)$

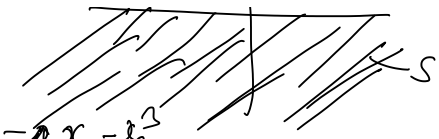
$f(x) = \frac{1}{2} x^T A x + bx + c$

$\nabla f(x) = \frac{1}{2}(A + A^T)x + b$

$\nabla^2 f(x) = \frac{1}{2}(A + A^T)$

$f: \mathbb{R}^n \to \mathbb{R}$

(4) $S = \{x \in \mathbb{R}^2 : x_2 < 0\}$



$f(x) = \frac{1}{2}x_1^2 + x_1x_2 + 2x_2^2 - 4x_1 - 4x_2 - x_2^3$

$Df(x) = \begin{bmatrix} x_1 + x_2 - 4 \\ x_1 + 4x_2 - 4 - 3x_2^2 \end{bmatrix}$

$D^2f(x) = \begin{bmatrix} 1 & 1 \\ 1 & 4 - 6x_2 \end{bmatrix}$

if $x_2 < 0 \Rightarrow$ both the eigenvalues are positive.

$\Rightarrow D^2f(x)$ is PD $\forall x \in S$

$\Rightarrow f$ is strictly convex

(5) $f(x) = 1/x, \quad x > 0$

$f'(x) = -1/x^2$

$f''(x) = \frac{2}{x^3}$

f is strictly convex

there is no m s.t. $\frac{2}{x^3} \geq m \quad \forall x > 0$

$\Rightarrow f$ is not strongly convex.

- [Equation] and [Equation]. Define [Equation] as [Equation]

If [Equation] s are convex for all [Equation] then [Equation] is also convex. $\quad -w_i f(x_i), \; w_i \geq 0$

if f_i's are concave for all i then h is also concave.

Example (1) $f(x) = x^2 + 2e^x$ is convex $\qquad f(x) = \sqrt{x}$

$\Rightarrow f'(x) = \frac{1}{2\sqrt{x}}$

(2) $f(x) = 3\sqrt{x} + \log(1+x) \quad x > 0$ $\qquad \Rightarrow f''(x) = -\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{x^{3/2}} < 0$

is concave

(3) $f(x) = x^2 - 2e^x$ convex or not?

— cannot conclude based on the above fact.

$\underbrace{\sum_a g_a f_a(x)}_{\text{convex}}$ $\qquad g_a > 0 \; \forall a$

convex

$g(\alpha) > 0$ for some α

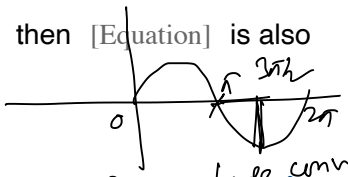
$g(\alpha) < 0$ for some α

- [Equation] and [Equation]. Define [Equation] as [Equation]

If [Equation] are convex in [Equation] for all [Equation] then [Equation] is also convex.

Example $\quad h(x) = \int_0^{2\pi} \sin\alpha \, e^{\alpha x} \, d\alpha$

convex a



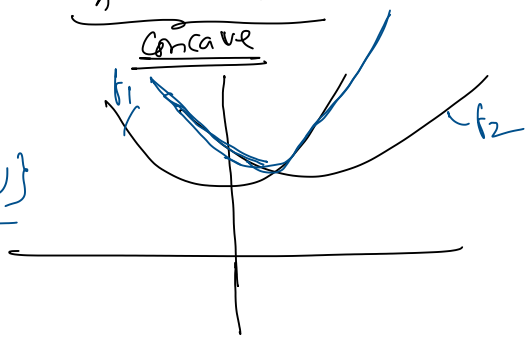
cannot conclude whether f is convex?

$g(x) = \int_0^{\pi/2} \sin\alpha \, e^{\alpha x} \, dx \Rightarrow g$ is convex.

$$h(x) = \underbrace{\int_{\pi}^{3\pi/2} \sin\alpha \; \underbrace{e^{\alpha x}}_{\text{convex}} \, d\alpha}_{\text{concave}}$$

Example

$f(x) = \max\{f_1(x), f_2(x)\}$



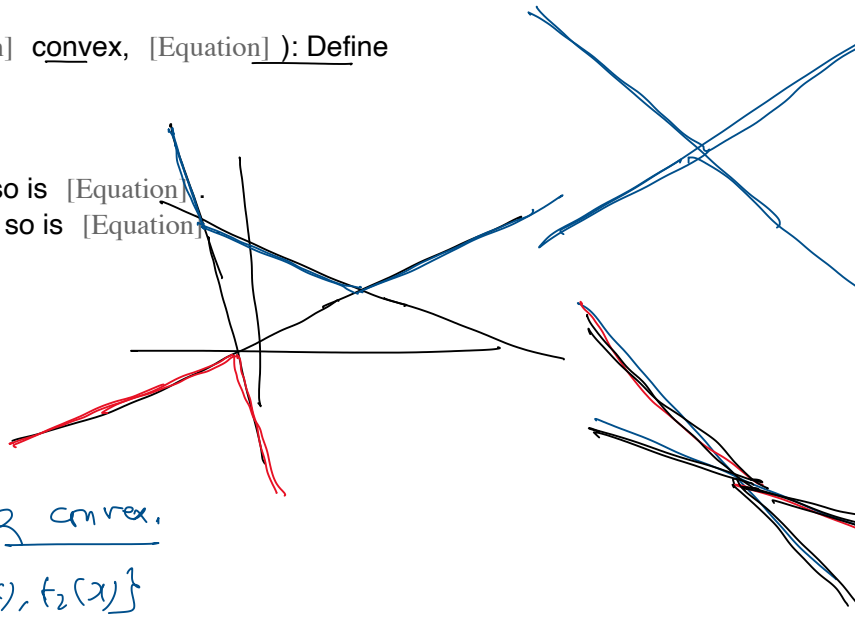
Maximum

~~Minimum~~ of convex functions ([Equation] convex, [Equation]): Define [Equation] as

[Equation]

[Equation]

- If [Equation] are convex functions so is [Equation].
- If [Equation] are concave functions so is [Equation]



Suppose $f_1, f_2 : S \to \mathbb{R}$ convex.

$f(x) = \max\{f_1(x), f_2(x)\}$

$$f(\alpha x + (1-\alpha)y) = \max\{f_1(\alpha x + (1-\alpha)y), f_2(\alpha x + (1-\alpha)y)\}$$

$$\leq \max\{\alpha f_1(x) + (1-\alpha) f_1(y), \alpha f_2(x) + (1-\alpha) f_2(y)\}$$

$$\leq \{\alpha \max\{f_1(x), f_2(x)\} + (1-\alpha)\max\{f_1(y), f_2(y)\}, \alpha \max\{f_1(x), f_2(x)\} + (1-\alpha)\max\{f_1(y), f_2(y)\}\}$$

$$= \alpha f(x) + (1-\alpha) f(y)$$

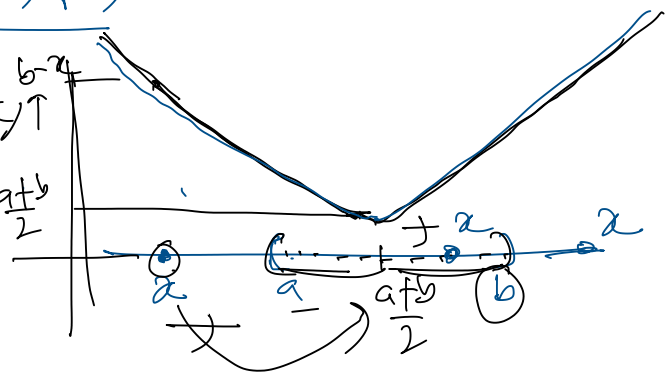
$\Rightarrow f$ is convex.

Example

① $C = [a, b]$

$f(x) = \max_{y \in C} |y - x|$

convex fn of y



$=$

$$f(x) = \max_{y \in C} |y - x|$$

$$f(x) = \begin{cases} b - x & \text{if } x < \frac{a+b}{2} \\ x - a & \text{if } x \geq \frac{a+b}{2} \end{cases}$$